H2O.ai Agents AI

From Theory to Practice

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Agenda

Goal

Gain an understanding of AI agents and how to effectively implement real-world use cases in h2oGPTe.

Audience

AI enthusiasts interested in AI agents as well as enterprise technology leaders and business stakeholders who need to understand and implement h2oGPTe agents in their organisations.

1. Intro to Agentic AI

Introduction, Spectra of AI Agents, Tool Use

2. h2oGPTe Agents

Features, GAIA Benchmark

3. Demos & Use Cases

Industry specific demos

4. AI Agents in Practice

Advice and guidance for implementing AI Agents in practice

Introduction to AI Agents

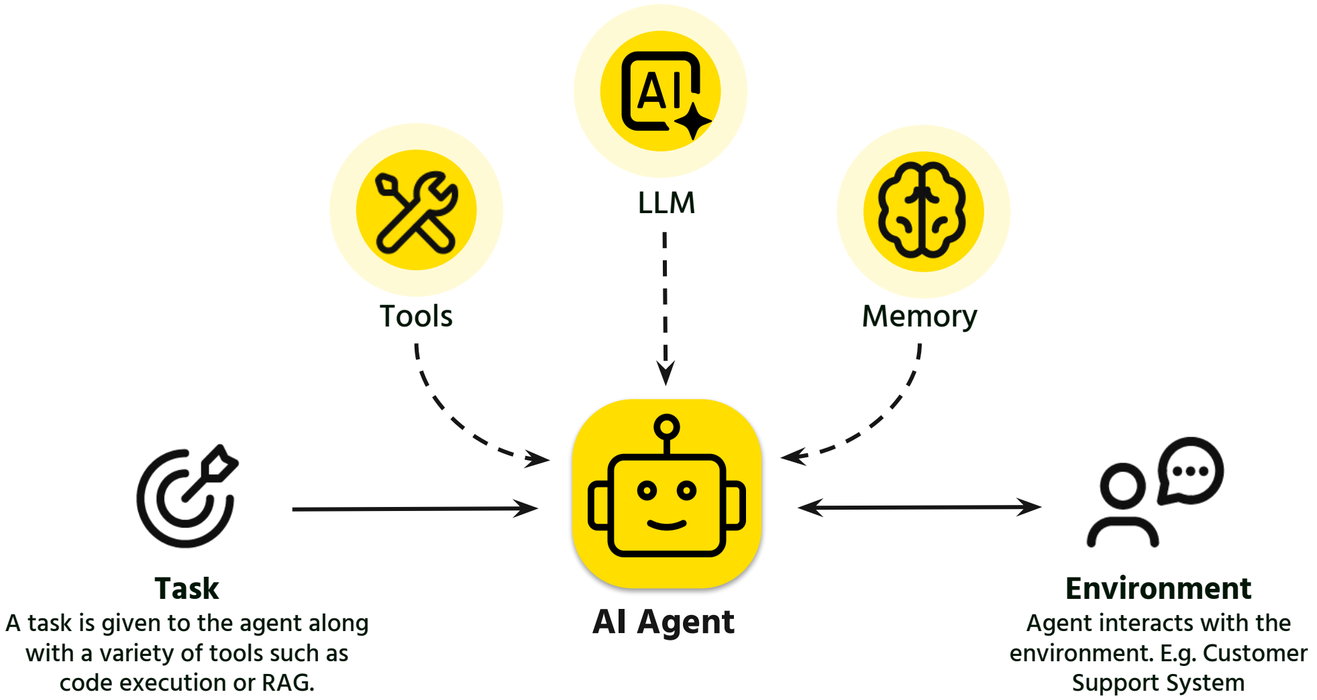
Agentic AI

Agentic AI actively **interacts with systems**, using tools or taking actions to achieve specific goals.

VS

Generative AI

Generative AI focuses on **producing content** like text, images, or code based on learned patterns. **RAG enhances** this by retrieving external information for context.



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Tools or Functions

Functionality that can accomplish a specific task given a set of inputs and are interfaces that an agent is "aware of" and can access and use to interact with.

Tools Available with h2oGPTe:

- Python and Bash code execution
- Web browsing
- Web search via Google or Bing
- Multi-modal understanding (text, images, audio) via OCR, caption, vision, and transcription models
- File handling across various formats
- Data science modeling and forecasting (including with DriverlessAI)
- Image generation
- Wolfram Alpha search
- Wikipedia search
- Multi-file code editing via Aider
- Advanced reasoning via OpenAI o1 or Qwen QwQ

Large Language Model (LLM)

An LLM serves as the “brain” of the AI agent, coordinating between understanding user needs, planning solutions, using tools and communicating effectively.

Frontier models are often preferred for building agents:

- Claude 3.7 Sonnet
- GPT-4o
- Deepseek R1
- GPT-o1
- Qwen 2.5 72B
- Meta LLama 3.3 70b
- Mistral Large Instruct

Memory

Memory allows the agent to 'learn' and adapt over time by utilising past interactions to inform current actions.

Memory is often implemented in the form of a Vector Database:

- h2oGPTe can be used to store and retrieve memory
- Chroma
- Pinecone
- Qdrant
- FAISS
- mem0

Introduction to AI Agents

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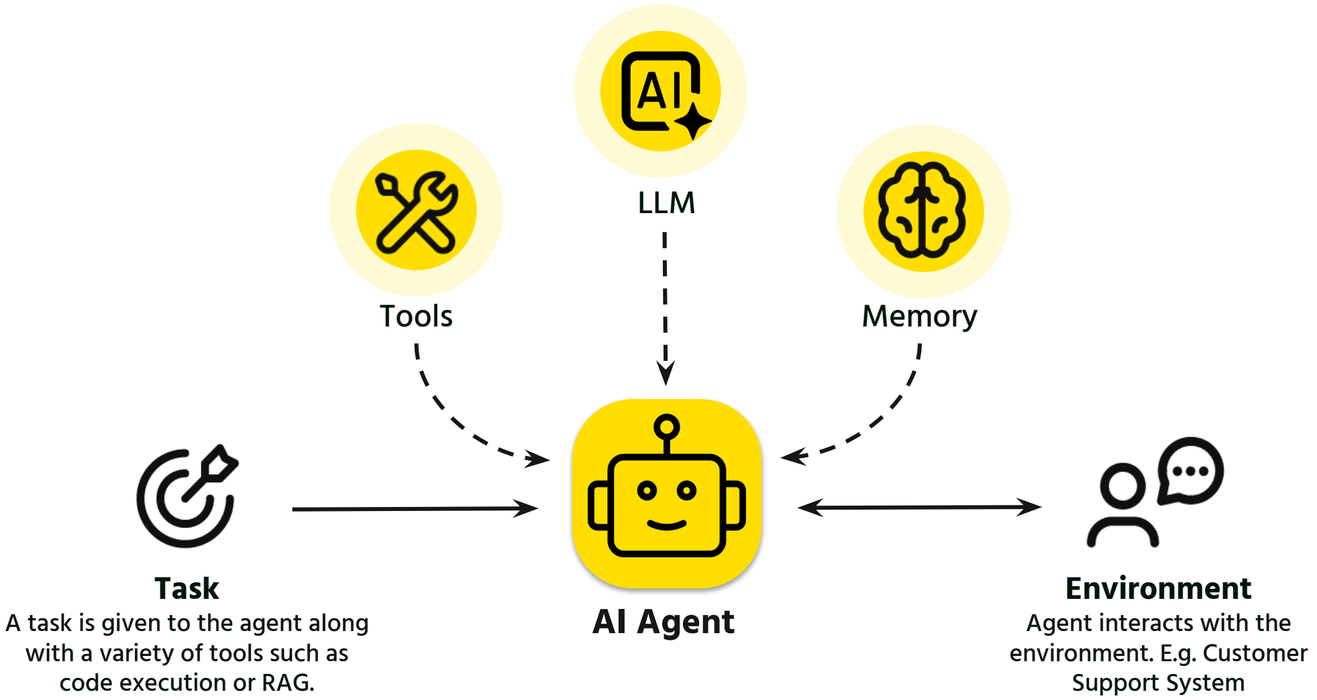

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Spectra of Agentic AI

No Agency

Simple LLM Calls
Pure Text Response

Limited Agency

Tool Use
Predefined Actions

Supervised Agency

Action Planning
Human Oversight

Full Agency

Autonomous Execution
Self-Determined Completion

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Spectra of Agentic AI

Enterprise h2oGPTe

No Agency	Limited Agency	Supervised Agency	Full Agency
Simple LLM Calls Pure Text Response	Tool Use Predefined Actions	Action Planning Human Oversight	Autonomous Execution Self-Determined Completion

ENTERPRISE GENERATIVE AI

h2oGPTe Agentic AI converges generative AI and predictive with purpose-built SLMs

The industry's **first multi-agent Generative AI platform** combines code-first agents with planner, memory, and execution modules, enabling tool calling and customizable tools (coming soon), while integrating the **strengths of Generative and Predictive AI** with air-gapped, on-premise deployment options.

Document AI with multimodal guided JSON generation

Multimodal audio & vision analysis

Coding assistant. Rapid prototyping & development

Autonomous agentic AI: execute multi-step workflows autonomously

Citation-based verification for transparent Retrieval Augmented Generation (RAG)

Customizable guardrails for AI safety

Intelligent model routing: optimal model selection for every task

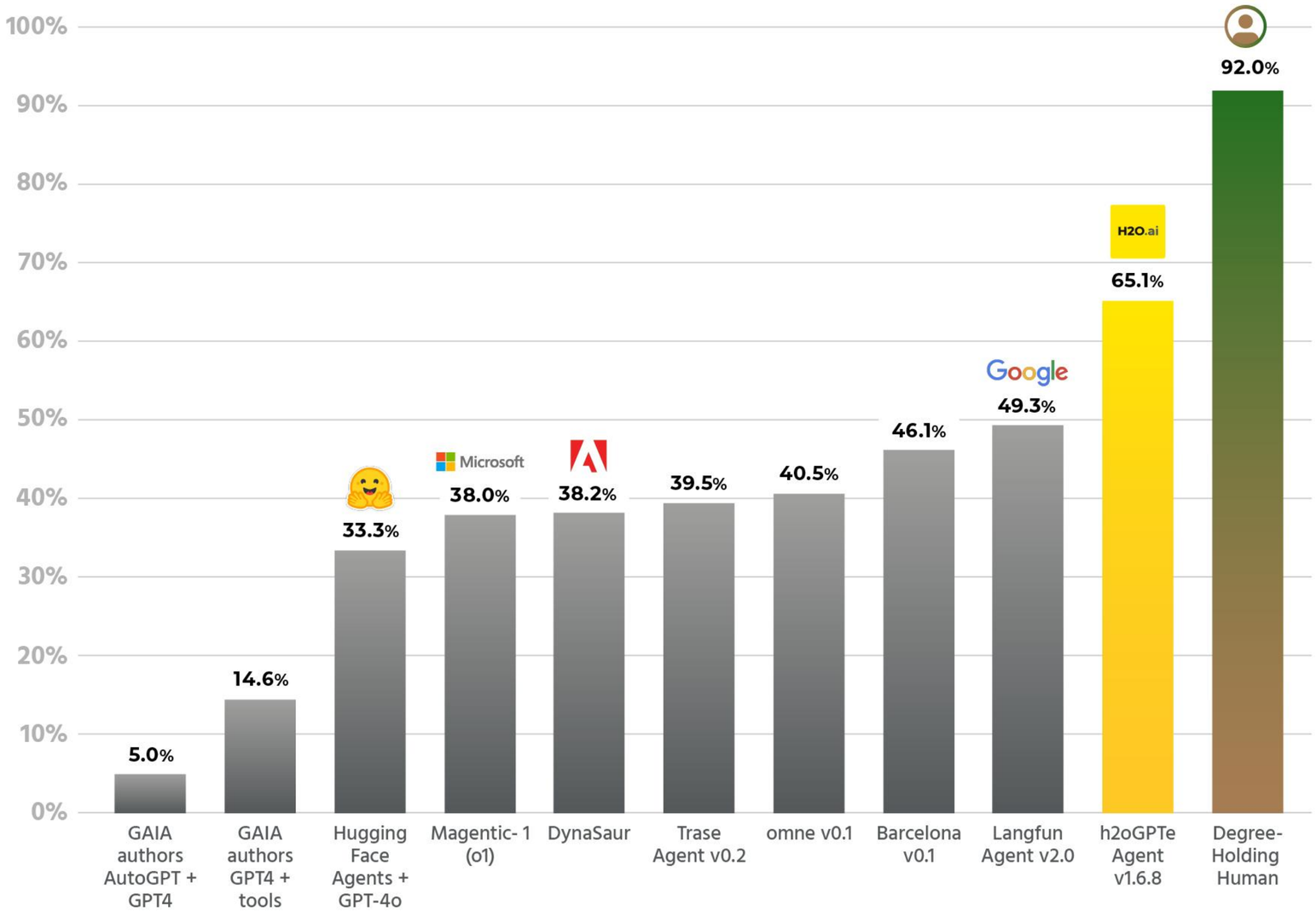
Model risk management for enhanced compliance and interpretability

The best of Generative & Predictive AI: Run AutoML using Agents

H2O.ai h2oGPTe: The World's Best Agentic AI Sets GAIA Benchmark Record

H2O.ai Takes the Top Spot in GAIA Benchmark: Surpassing Google by a Stunning 15.8% Margin

H2O.ai h2oGPTe Agentic AI Tops GAIA Dec 2024



Source: https://huggingface.co/spaces/gaia-benchmark/leaderboard

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Enterprise h2oGPTe

Agents Demo

AI Financial Analyst Workflow

Investment Analysis for H2O.ai

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TOOLS

Google Search

Internet Access

Phase 1: Market Intelligence

Analyse market trends and financial reports

TOOLS

"Ask questions about documents"

Phase 2: Compliance Check

Review regulations & compliance requirements

TOOLS

Code Execution

Advanced Reasoning

Phase 3: Risk Assessment

Calculate exposure & stress test scenarios

Phase 4: Investment Decision Report

Tools: Python coding

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AI Government AI Policy Workflow

Develop a National AI Skills Development Framework

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TOOLS

Google Search

Internet Access

Phase 1: Policy Research

Review existing policies & legal frameworks

TOOLS

"Ask questions about documents"

Phase 2: Public Impact

Access service delivery & community accessibility

TOOLS

Code Execution

Advanced Reasoning

Phase 3: Impact Assessment

Calculate economic & social outcomes

Phase 4: Policy Decision Report

Tools: Python coding

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AI Medical Diagnosis Researcher Workflow

Evidence-based Research Assistant

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TOOLS

Google Search

Scholar Papers Search

Phase 1: Literature Search

Search latest research & systematic reviews

TOOLS

"Ask questions about documents"

Phase 2: Evidence Analysis

Evaluate quality & relevance of research findings

TOOLS

Code Execution

Advanced Reasoning

Phase 3: Clinical Synthesis

Synthesise research into clinical recommendations

Phase 4: Diagnosis Report Generation

Tools: Python coding

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Telco AI Customer Retention Analyst

Data Analysis & Modelling Agent

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Churn Data

Customer Profile

- Demographics
- Tenure & History

Services

- Current Subscriptions
- Usage Patterns

Financial Data

- Monthly Charges
- Payment History

h2oGPTe Agents

Python Coding
Data pre-processing & analysis.

H2O Driverless AI
Churn Modelling using AutoML and interpretability.

Advanced Reasoning
Open AI's GPT-o1 for planning and validating methodology.

Insights

Visualisations

- Churn contribution by segment
- Feature importance

Key Findings

Churn rate breakdown per demographic, service and financial factors

Recommendations

Recommended intervention for reducing churn based on analysis.

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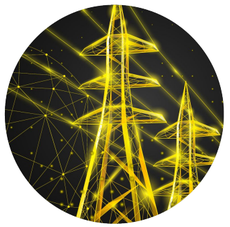

AI Power Plant Performance Analyst

Data Analysis & Modelling Agent

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Power Plant Data

Sensor Inputs

- Temperature
- Ambient Pressure
- Relative Humidity
- Exhaust Vacuum

Target Variable

Net hourly electrical energy output.

h2oGPTe Agents

Python Coding
Data pre-processing & analysis.

H2O Driverless AI
Churn Modelling using AutoML and interpretability.

Advanced Reasoning
Open AI's GPT-o1 for planning and validating methodology.

Insights

Visualisations

- Effect of each ambient variable on power output
- Seasonality and trends

DAI Model

Machine Learning Model artifact to forecast hourly power output.

Recommendations

Suggest optimal operating parameters, alert thresholds and control system adjustments.

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AI Agents in Practice
Caveats & Tips

Data Integration & Access
Core systems and databases may not be easily accessible via API. May need heavy engineering support to uplift legacy systems.

SME Oversight
Engage subject matter experts (SMEs) throughout development to ensure alignment with regulations and internal procedures.

Implementing Guardrails
Establish input and output guardrails to help ensure the AI agent acts within defined constraints This may be important for regulatory compliance.

Testing & Validation
Implement red teaming strategies and continuously monitor the behaviour of your AI agent through traces.

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Recap

Agentic vs Generative AI

Agentic AI actively interacts with systems and takes autonomous actions, while generative AI focuses on content creation.

h2oGPTe Capabilities

h2oGPTe sets new benchmarks in agentic AI performance, featuring advanced tool calling, multimodal processing, and enterprise-grade guardrails for deployment.

h2oGPTe Use Cases

Researcher and coding agents in h2oGPTe can decrease the time to discover useful insights and recommendations whilst remaining transparent in its thought process.

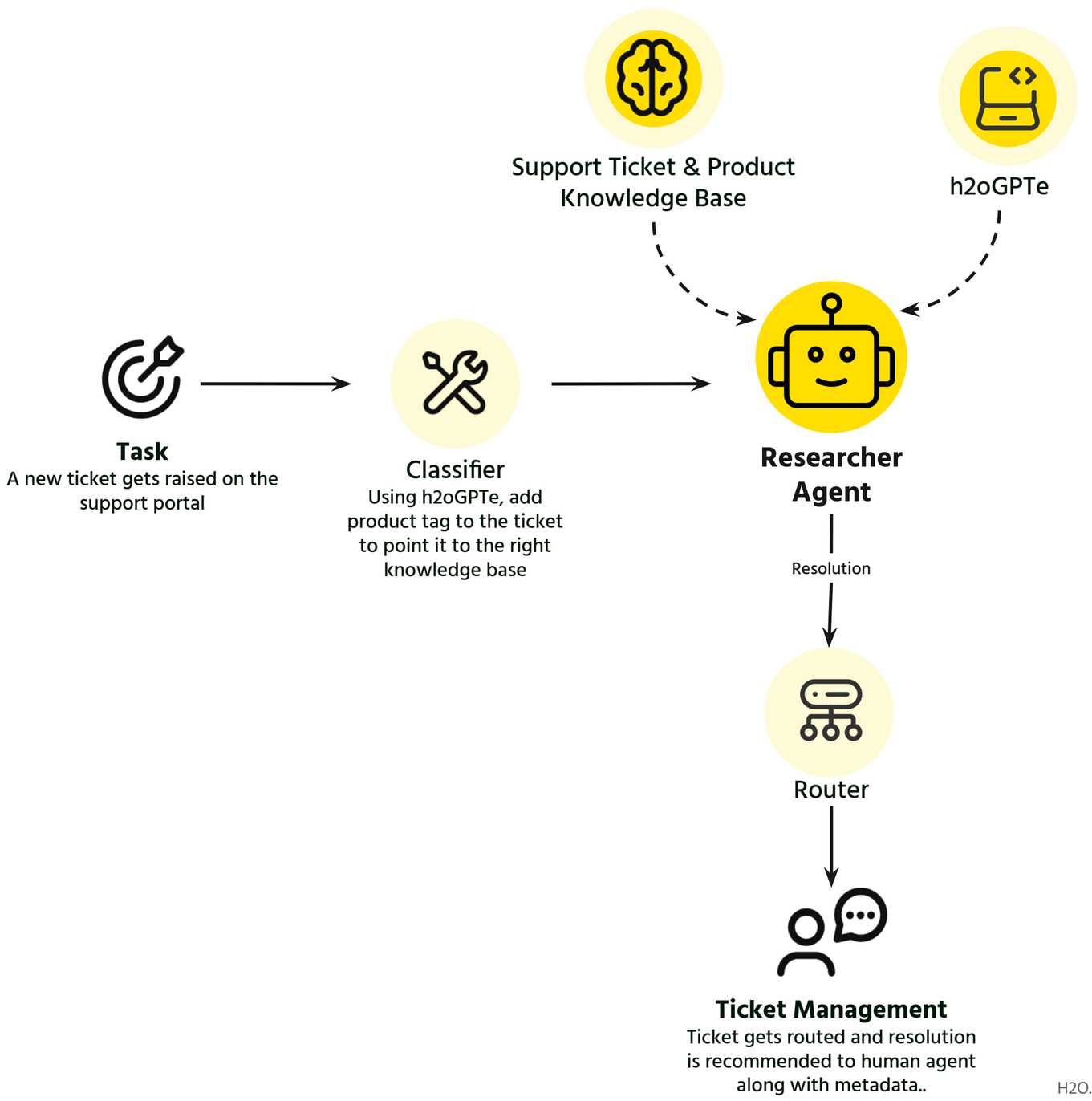
Your turn!

Keep learning & building your skills on h2o.ai/university

Case Study 1

Agentic AI to autonomously support ticket management processes for faster resolution

The agent, powered by H2OGPTe capabilities, executes a seamless workflow that includes sequential tasks: ticket tagging, routing, and suggesting recommendations. It leverages historical ticket data, conversations, and product documentation as the knowledge base. This approach enhances efficiency, improves accuracy, and adapts over time.



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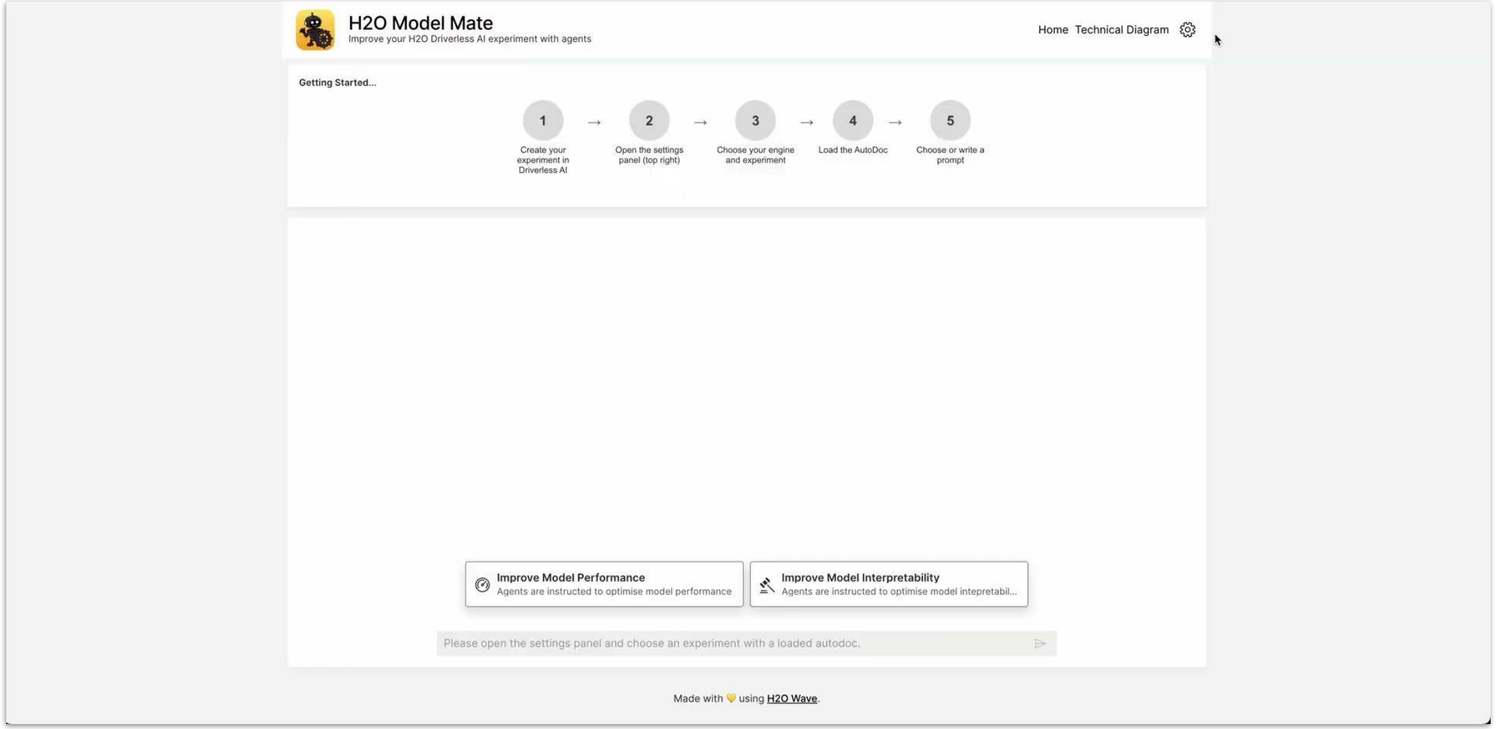
Case Study 2

H2O Model Mate

An intelligent assistant that helps you improve your H2O Driverless AI models using AI agents. The application provides guided suggestions and automated improvements for model optimization, interpretability, and custom objectives.



Part 2: Whiteboard Workshop

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Agentic AI Use Case Ideas

Post-Call Action Orchestrator

The agent automatically captures commitments and required actions from customer service calls, creates structured work items, assigns them to appropriate teams, and monitors their completion while keeping customers updated on progress through their preferred communication channels.

Document Processing Automation

An AI agent manages the end-to-end processing of standard banking documents like loan applications and account openings, intelligently routing documents to appropriate departments, following up on missing information, and maintaining audit trails of all actions taken.

Intelligent Customer Service Routing

The agent evaluates customer inquiries in real-time, considering factors like urgency, complexity, and customer history to route queries to the most appropriate service channel or representative, while proactively gathering relevant information to expedite resolution.

Regulatory Compliance Monitor

An AI agent continuously monitors transaction patterns and account activities, identifying potential compliance issues and generating detailed reports for review. It adapts its monitoring parameters based on feedback from compliance officers and regulatory updates.

Account Maintenance Assistant

The agent proactively identifies and resolves routine account maintenance issues, such as updating expired card information, managing recurring payments, and alerting customers about potential service disruptions before they occur.

Financial Education Coach

An AI agent provides personalized financial education by analyzing a customer's banking behavior and knowledge level, then delivering targeted educational content and practical exercises to improve their financial literacy and decision-making skills.

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ACTIVITY 1

30 mins

Brainstorming & Prioritisation

Whiteboarding Template

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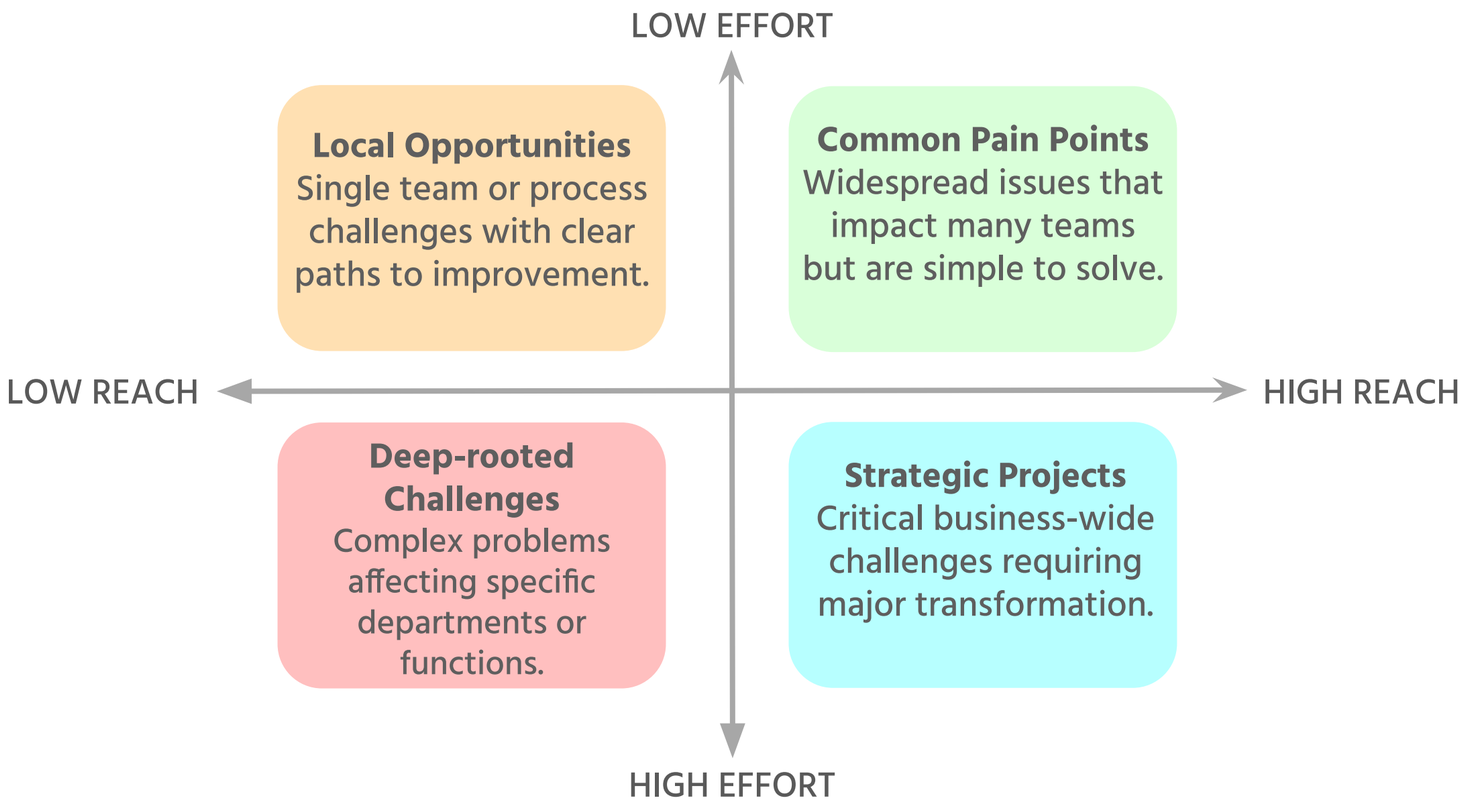
01 Identify **pain points, business goals** and other ideas.

02 **Rank ideas** using a prioritisation matrix.

03 Focus on **low effort, high reach** use cases.



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ACTIVITY 2

30 mins

Current State
Whiteboarding Template

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01 Describe the **motivation** for the use case idea.

02 Identify **key stakeholders** or process owners.

03 Describe the **expected impact** from solving the problem.

04 Describe the solution that is **currently implemented.**

Use Case Motivation	Key Stakeholders	Expected Impact	Current Implementation
Repetitive customer queries consume agent time that could be spent on complex issues	Support Agents and Training Managers	More efficient use of agent expertise, allowing them to focus on high-value customer interactions	Customers initially interact with a chatbot that tries to answer simple questions.
...	...	...	

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ACTIVITY 3

30 mins

Opportunity Areas
Whiteboarding Template

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01 Describe how **agents** might be leveraged to **solve** pain points

02 Define ROI based on **business impact**

03 Identify any **system integration** required

04 Define any input and output **guardrails**

Agentic Solution	Business Impact	System Integration	Guardrails
Deploy an autonomous agent that analyzes incoming queries, automatically resolves common issues through system actions, and only routes complex cases to human agents.	Reduction in average handling time for common queries.	Integration with ticketing system and relevant backend systems that can execute common customer requests (e.g., password resets, balance inquiries, statement requests).	Strict verification of customer identity before any action, mandatory human review for any system changes.
...	...	...	...

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Recap & Conclusion

Agentic vs Generative AI

Agentic AI actively interacts with systems and takes autonomous actions, while generative AI focuses on content creation.

h2oGPTe Capabilities

h2oGPTe sets new benchmarks in agentic AI performance, featuring advanced tool calling, multimodal processing, and enterprise-grade guardrails for deployment.

Workshop Outcomes

Through collaborative whiteboarding sessions, we identified key pain points, prioritized use cases based on ROI and effort, and developed concrete implementation strategies.

Next Steps

Schedule follow-up meetings to refine prioritized use cases.

Appendix -
Contact Centers + Agentic

Key Ideas : Agentic AI for Contact Centers

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Automate customer conversation

Automate during-call action items

- Conversations [Data]
- Meta Data [Documents]
  - Intent Identification

Agentic Actions
  - Conversation Intent Validation + Metadata Matching
  - Action Identification
  - API / Function call identification
  - Executing the Function call (ie. automated actions)
    - Database Update
    - CRM update
  - Notifications, Logging

Automated Quality Assurance

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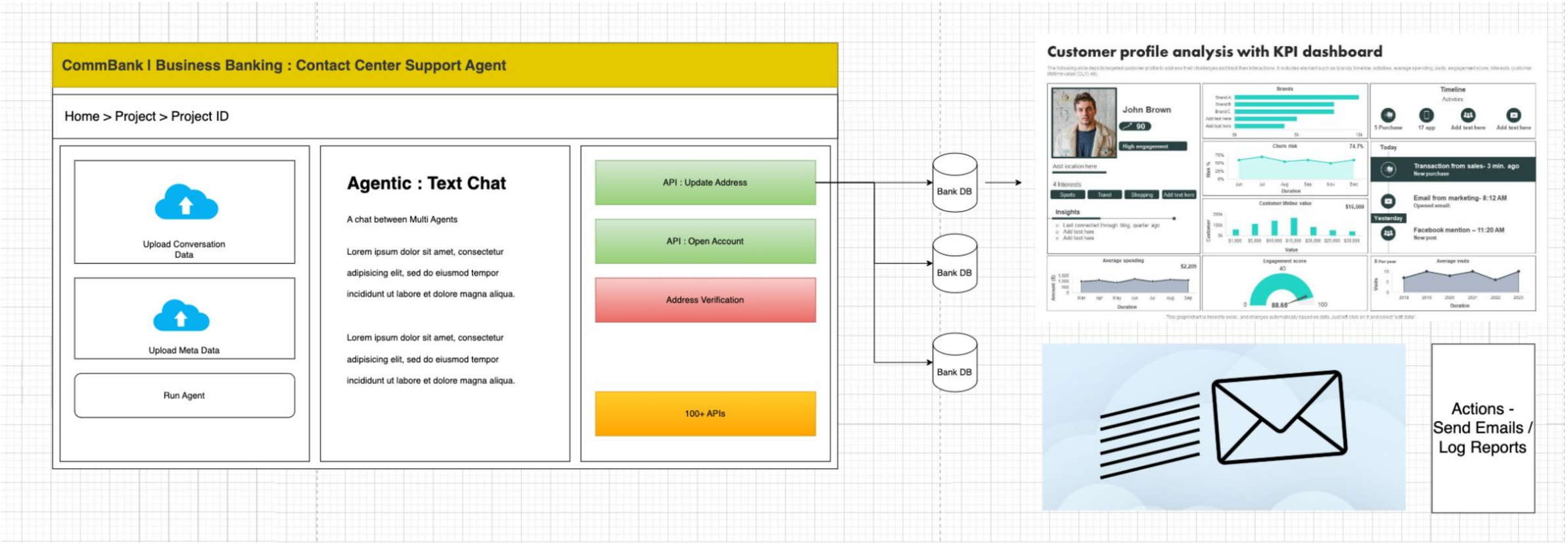
CommBank I Business Banking : Contact Center Support Agent
Home > Project > Project ID
Upload Conversation Data
Upload Meta Data
Run Agent
Agentic : Text Chat
A chat between Multi Agents
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API : Update Address
API : Open Account
Address Verification
100+ APIs
Bank DB
Bank DB
Bank DB
Customer profile analysis with KPI dashboard
John Brown
Actions - Send Emails / Log Reports

Appendix

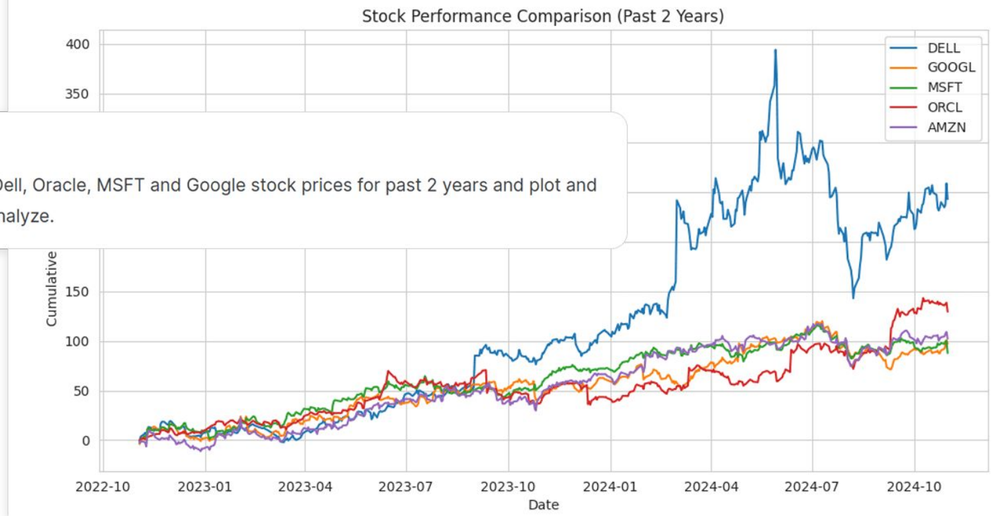
Autonomous agentic AI: execute multi-step workflows autonomously

h2oGPTe Agents bring true autonomy to your workflows, enabling LLMs to handle multi-step tasks like web research, data modeling, database access, and iterative code execution. Programmatic and continuous, these agents reduce manual workload by executing tasks requiring sequential logic, real-time decision-making, and data handling.

h2oGPTe Agents can autonomously generate multi-page PDFs with charts, tables, and flowcharts based on real-time data—complete with source code for full transparency.

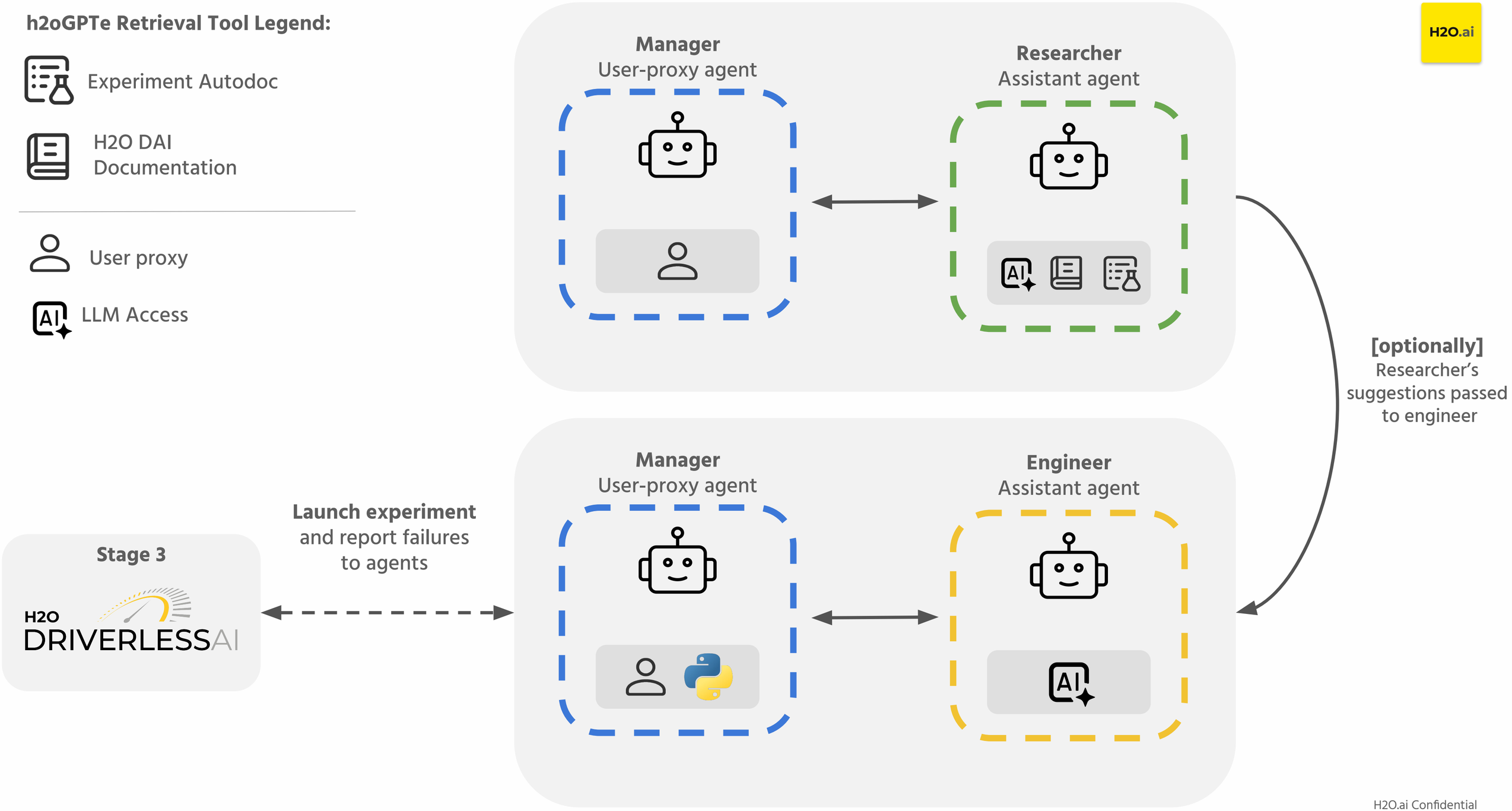
You

Download Dell, Oracle, MSFT and Google stock prices for past 2 years and plot and compare, analyze.



Key Insights:

1. Dell's Exceptional Performance: Dell significantly outperformed all other stocks, reaching over 350% cumulative return at its peak around July 2024. This suggests a remarkable turnaround or successful strategic shift for the company.
2. Oracle's Strong Position: Oracle emerged as the second-best performer, showing a late surge in 2024 and outperforming the other tech giants in the latter part of the year.
3. Tech Giants' Similar Trajectories: Amazon, Google, and Microsoft showed similar moderate growth patterns, generally staying within the 50-150% cumulative return range.
4. Sector-wide Trends:
   - Early 2023 Dip: A noticeable dip occurred for most stocks around January 2023, followed by recovery and sustained growth.
   - Increased Volatility in 2024: The period from April 2024 to October 2024 showed increased volatility for all stocks, with Dell experiencing the most dramatic fluctuations.
5. Overall Growth: All five companies demonstrated positive and significant growth over the two-year period, indicating the strength and resilience of the tech sector.



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Agenda (draft)

1. Goals (5 mins)
   a. Workshop purpose
2. h2oGPTe Agents Demo (30 mins)
3. Agentic Apps (30 mins)
4. Domain Specific Case Study Discussion (60 mins)
   a. e.g. Business Banking Contact Centre Flow
      i. Current State (CBA to prefill template)
      ii. Identify areas for agents
      iii. Metrics for success (domain specific e.g. NPS)
      iv. Prioritisation (ROI vs Complexity)

1. Goals: Identify and prioritise use case ideas where agents can deliver the greatest ROI.
2. h2oGPTe Agents demo
    a. GAIA benchmark results
    b. Tool calling e.g. RAG Text, Web Search, Audio-Video Transcription
3. Agentic Apps
    a. H2O Model Mate
    b. Telecom Help Desk
4. Domain Specific Case Study Whiteboarding
    a. Describe current state
        i. Identify key pain points in decisioning processes
        ii. Identify key stakeholders or process owners
        iii. Describe the expected impact from solving the problem
    b. Identify areas where agents can be applied
        i. Describe how agents might be leveraged to solve pain points
        ii. Define ROI based on business metrics (e.g. Improve identification and rectification of compliance lapses by X%)
        iii. Identify any system integration required
        iv. Define guard rails
    c. Prioritise
        i. Prioritisation matrix to rank use cases based on ROI (low effort, high value first)
        ii. Define clear owners and next steps

ACTIVITY 2

30 mins

Prioritisation Whiteboarding Template

01 **Prioritisation matrix** to rank use cases based on ROI.

02 Define clear **owners and next steps**

LOW EFFORT

Fill-ins
Simple automations that provide marginal ROI but have few guardrails and system integration is available.

Quick Wins
Minimal risk with few guardrails and system integration is available.

1. Assign ownership
2. Owners should arrange follow-up meetings to refine and expand use cases

LOW VALUE

HIGH VALUE

Time Sinks
Complex integrations and guardrails that solve difficult business problems or benefit few customers.

Strategic Projects
Complex integrations, guardrail and business problem but delivers substantial ROI.

HIGH EFFORT

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ACTIVITY 2

30 mins

Current State Whiteboarding Template

Identify the Motivation

01 Identify key **pain points** in decisioning processes
Wish list - ideas ?
Business Goals

02 Identify **key stakeholders** or process owners

03 Describe the **expected impact** from solving the problem

Pain Point	Key Stakeholders	Expected Impact
Repetitive customer queries consume agent time that could be spent on complex issues	Support Agents and Training Managers	More efficient use of agent expertise, allowing them to focus on high-value customer interactions
...	...	...

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Demo ideas:

- Financial services
  - Investment Analysis
  - Risk Assessment Predictive Modeling
  - Regulatory compliance real-time monitoring
  - Portfolio optimisation market analysis
  - fraud detection pattern recognition
- Government
  - Policy Analysis document processing
  - public service optimiser
  - emergency response real-time coordination
  - grant management application processing
  - regulatory updates compliance tracking
- Telco
  - network performance real time monitor
  - customer support
  - infrastructure planning
  - service quality automated assessment
  - resource optimisation load balancing
- Energy
  - grid management real time control
  - predictive maintenance equipment monitoring
  - environmental impact emissions tracking
  - resource planning demand forecasting
  - safety protocols automated compliance
- Healthcare
  - Clinical decision support evidence based analysis
  - patient care optimiser
  - research analysis literature review
  - equipment management predictive maintenance
  - compliance management regulatory tracker

Finally, reveal all use cases developed by AI

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